

RESPONSE TO REVIEWER COMMENTS

Manuscript Title: *Correlation Mapping of Sanitation Access and Public Health Outcomes in Tamil Nadu*

We sincerely thank the reviewer for the valuable and constructive comments. The suggestions have significantly improved the technical quality, clarity, and academic rigor of the manuscript. All comments have been carefully addressed and corresponding revisions have been incorporated in the revised version. A detailed point-by-point response is provided below.

General Comment

Comment:

The paper is important and policy-relevant but requires technical and presentation improvements to meet academic standards.

Response:

We appreciate this positive assessment. The manuscript has been thoroughly revised to enhance methodological clarity, statistical reporting, formatting consistency, and overall presentation quality. Regression results, confidence intervals, limitations, and modeling explanations have now been included.

Specific Comments and Responses

1. Include proper regression results with confidence intervals, R^2 values, and effect sizes.

Response:

We have now added a detailed regression analysis section including:

- Multiple linear regression model equation
- Coefficients with standard errors
- 95% confidence intervals
- R^2 and adjusted R^2 values
- F-statistic and significance level

A regression results table has been added to improve clarity and statistical transparency.

2. Explain why open defecation still exists despite 100% toilet coverage.

Response:

A new subsection titled “*Behavioral Gap in Sanitation Usage*” has been added. This section explains:

- Cultural norms and habit persistence
- Water availability constraints
- Maintenance and quality issues
- Social stigma and shared facility concerns
- Gaps between infrastructure provision and behavioral adoption

Relevant literature on behavior change and community-led sanitation has been incorporated.

3. Add a limitations section.**Response:**

A dedicated “*Limitations*” section has been added before the conclusion. It discusses:

- District-level data aggregation issues
- Survey self-reporting bias
- Time differences across datasets
- GIS boundary constraints
- Correlation vs causation limitation

4. Provide proper references for IMR and diarrheal disease data.**Response:**

All health indicators, including IMR and diarrheal disease statistics, are now properly cited using NFHS, IDSP, and government health department sources. References have been formatted consistently in IEEE style.

5. Differentiate between primary and secondary sources.**Response:**

The Methodology section now clearly distinguishes:

- **Primary sources:** NFHS, SBM Dashboard, Census, IDSP
- **Secondary sources:** WHO reports, published studies, news sources

This improves data transparency and credibility.

6. Expand discussion on women and adolescent girls.

Response:

The section on gender disparities has been expanded to discuss:

- School absenteeism due to lack of sanitation
- Menstrual hygiene challenges
- Safety risks and harassment
- Psychological stress
- Link between sanitation and female literacy

This strengthens the social equity dimension of the study.

7. Explain relationship between sanitation, education, and women's safety.

Response:

A conceptual linkage framework has been incorporated in the Discussion section:

Sanitation Access → School Attendance → Female Literacy → Health Awareness → Improved Health Outcomes

The interaction between literacy and sanitation benefits is also supported by regression findings.

8. Improve grammar and sentence structure.

Response:

The entire manuscript has been carefully proofread.

- Repetitive phrases removed
- Sentence structures improved
- Tense consistency ensured
- Technical writing refined

9. Use consistent terminology.

Response:

The manuscript now consistently uses the term “**Sanitation Coverage**” throughout the document to maintain uniformity.

10. Number all figures and tables correctly.

Response:

All figures and tables are now sequentially numbered and properly referenced in the text (e.g., Fig. 1, Table I).

11. Add proper captions and refer clearly in text.

Response:

Each figure and table now includes descriptive captions and is clearly cited in the corresponding sections.

12. Improve maps and charts (legends, scales, labels).

Response:

GIS maps have been updated to include:

- Legends
- Scale bars
- North arrows
- District labels
- Data source and year

This enhances interpretability and academic presentation.

13. Strengthen the conclusion linking findings to policy recommendations.

Response:

The Conclusion section has been rewritten to:

- Highlight key regression findings
- Emphasize statistical strength (R^2 value)
- Connect findings with district-level policy actions
- Reinforce behavioral intervention importance

14. Add future work section.

Response:

A new section titled “*Future Work*” has been added, suggesting:

- Longitudinal behavioral studies
- Climate change and sanitation linkage
- AI-based predictive sanitation modeling
- Village-level GIS studies

Detailed Comment 1: Paper not in format.**Response:**

The manuscript has been fully restructured following standard journal format:

- Abstract
- Keywords
- Introduction
- Literature Review
- Methodology
- Results
- Discussion
- Policy Implications
- Limitations
- Conclusion
- Future Work
- References

Detailed Comment 2: Need to explain modelling more clearly.**Response:**

The modelling framework has been expanded to include:

- Justification for Pearson correlation
- Justification for multiple regression
- Model equation
- Assumption explanation (linearity, multicollinearity, etc.)

- Statistical reporting standards

This significantly strengthens the technical rigor.

Final Statement

We are grateful for the reviewer's insightful feedback. The revisions have substantially improved the clarity, methodological robustness, and policy relevance of the manuscript. We hope the revised version meets the journal's academic standards and look forward to further feedback.